Beyond averages

Timestamp gaps & cancellation concentration

01

Typical ≠ extreme

02

Departure ≠ arrival

03

Dates ≠ averages

[Your name] • [Course] • [Presentation date]

One airport. One month. Three lenses.

22 September–22 October 2026

23,705

Flight-event records

31

Local service dates

11,560

Timing-eligible departures

Recorded timestamp gaps — not validated operational delays

What does an average hide?

1

Typical experience or extreme risk?

Compare morning and evening departures.

2

Later departure → later arrival?

Link the timestamps within each record.

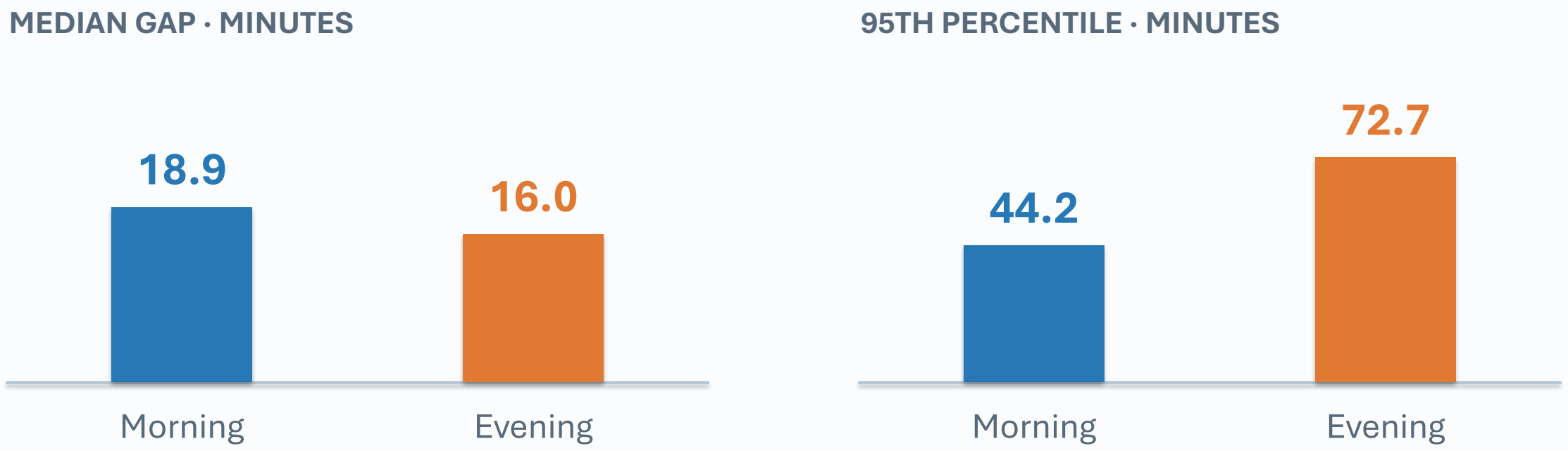
3

A monthly rate or a few unusual dates?

Measure cancellation concentration.

Evening: lower median, heavier upper tail

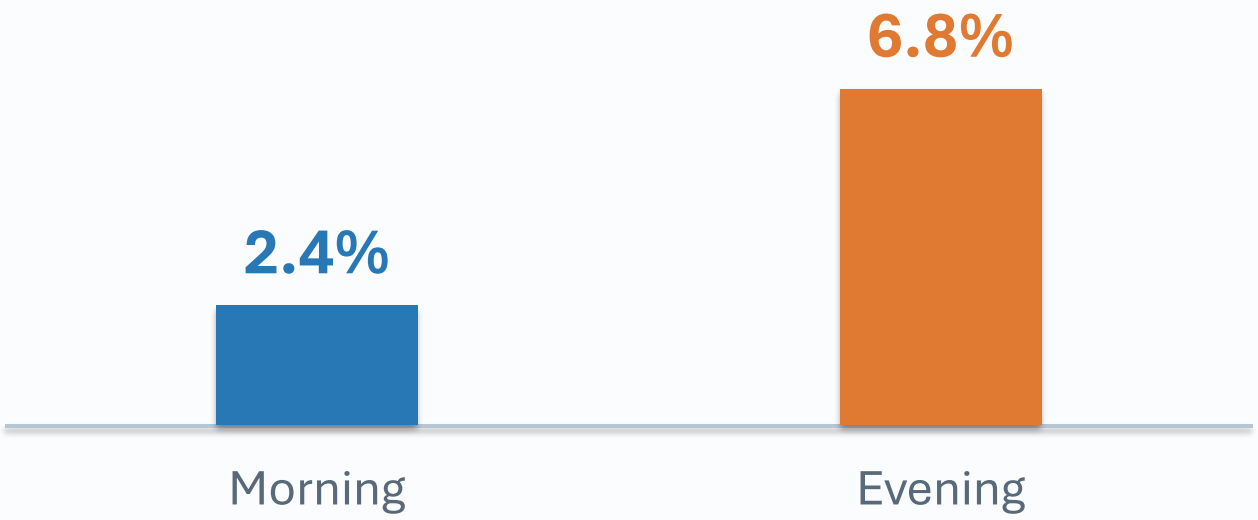
Morning 06:00–11:59 | Evening 17:00–22:59 • Shanghai local time



A smaller typical gap does not mean less extreme risk.

Extreme gaps are about 2.8× as common in evening

Extreme = recorded departure gap ≥60 minutes



2.8×

Observed fraction ratio

Unadjusted • not a causal effect

But could route mix or the date explain the pattern?

The tail difference persists within route and date

860 route-date cells • 31 dates • evening minus morning

Mean gap

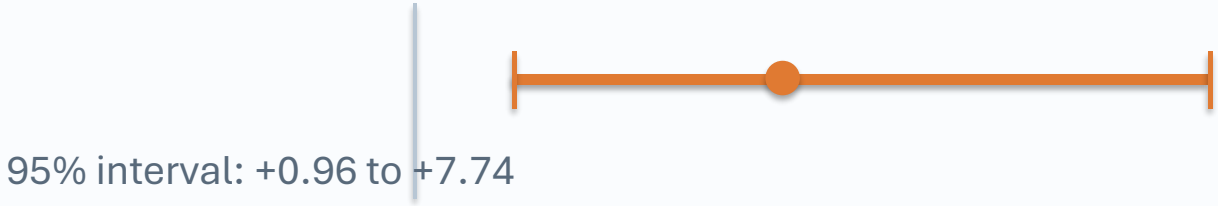
+0.52 minutes



95% interval: -2.31 to +4.71

Gap ≥60 fraction

+3.60 percentage points



95% interval: +0.96 to +7.74

Mean contrast ≈ zero; extreme-tail contrast stays positive.

A later departure need not mean later arrival

SHA departure cohort • paired complete records only

6,923

Departure gap ≥15 min

→

5,721

Arrival gap <15 min

82.6%

Conditional fraction

95% date-bootstrap interval: 78.7%–86.4%

This is a timestamp pattern — not proof of time made up in flight.

Before telling a story, check what the clocks measure

Schedule

What event does the published time describe?

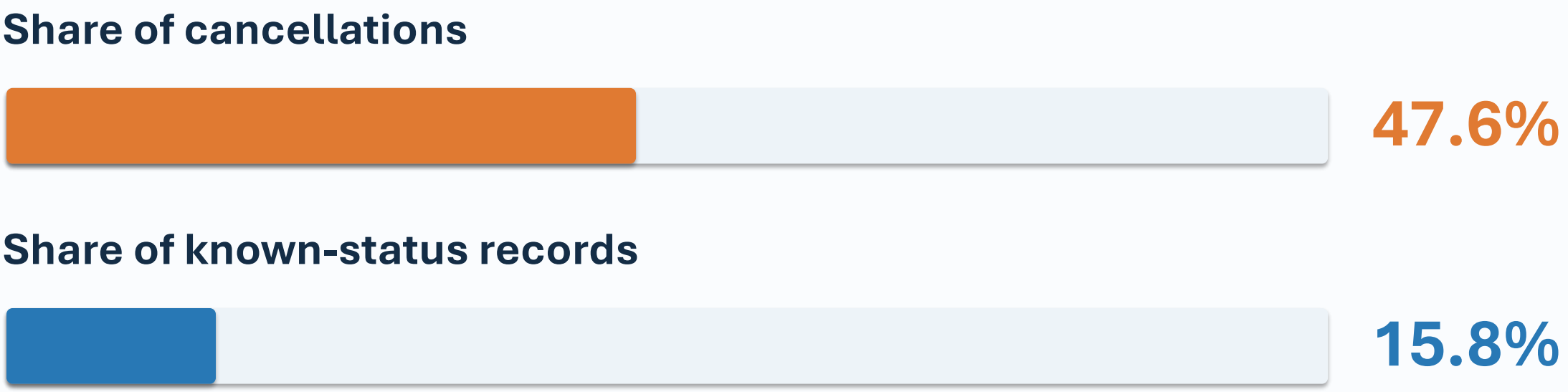
Recorded event

What event does the observed time describe?

Possible event mismatch ≠ proven aircraft recovery

Five dates account for almost half the cancellations

Top five dates selected by observed cancellation count

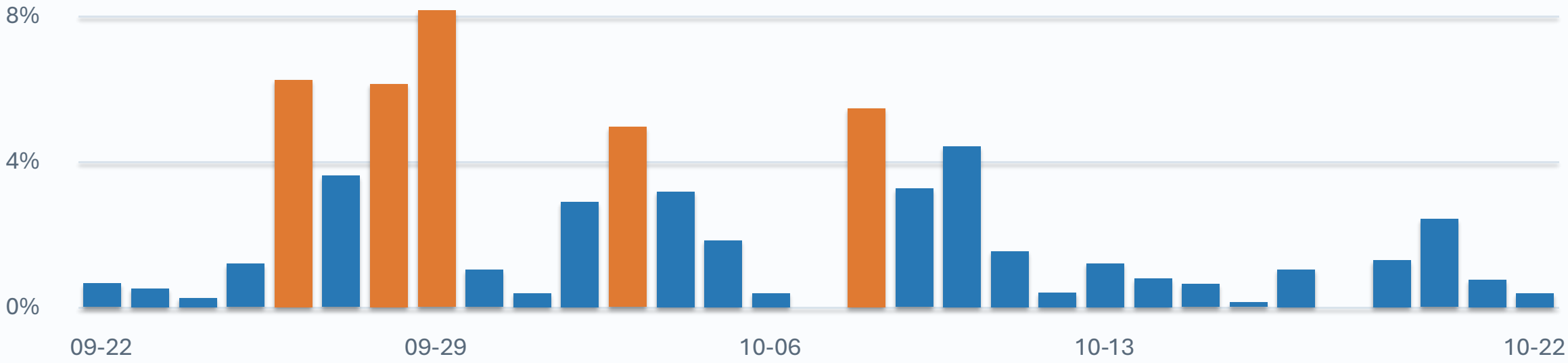


Cancellation rates: 6.19% on these dates vs 1.28% otherwise

The daily pattern is more revealing than one monthly rate

Canceled ÷ known-status event records • orange = top five dates by count

29 Sep: 62 / 761 canceled (8.15%)



Concentration is clear. Its cause is not identified.

Strong patterns. Limited claims.

01

Unverified clocks

Validate event definitions

02

One month, one airport

Replicate in another month

03

Exploratory, not causal

Predefine the next test

Do not claim airline rankings, holiday effects, or proven recovery.

Three takeaways to remember

1 Typical experience ≠ extreme-tail risk

2 Timestamp reduction ≠ proven recovery

3 Monthly average ≠ daily experience

Look beyond averages — and stay honest about measurement.

Methods & robustness at a glance

Timing sample	Known, noncanceled status + required timestamps
Matching	Same route and date; equal cell weights
Uncertainty	2,000 whole-date bootstrap samples
Sensitivity	≥2 records per window; leave one date out
Paired check	Identity-flag exclusion gives a similar fraction

Sources & reproducibility

Data	User-supplied SHA flight-event dataset (2026)
Clustered inference	Cameron & Miller (2015) • doi:10.3368/jhr.50.2.317
Bootstrap	Efron (1979) • doi:10.1214/aos/1176344552
Air-transport context	Mayer & Sinai (2003) • doi:10.1257/0002828033769206269

Full citations and speaking guidance are in the slide notes.